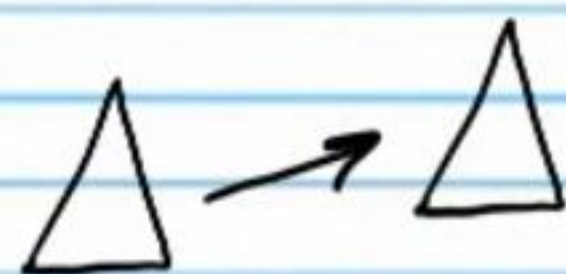


Transformations and the Library of Relations (Part II)

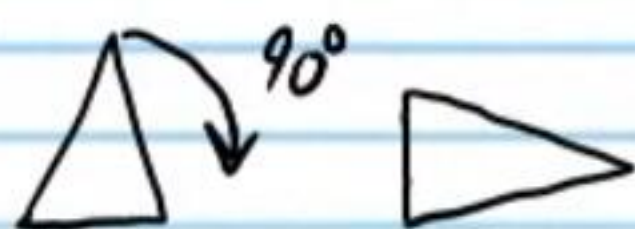
Review on Transformations

I) Translation.



II) Rotation.

* There must be a center of rotation



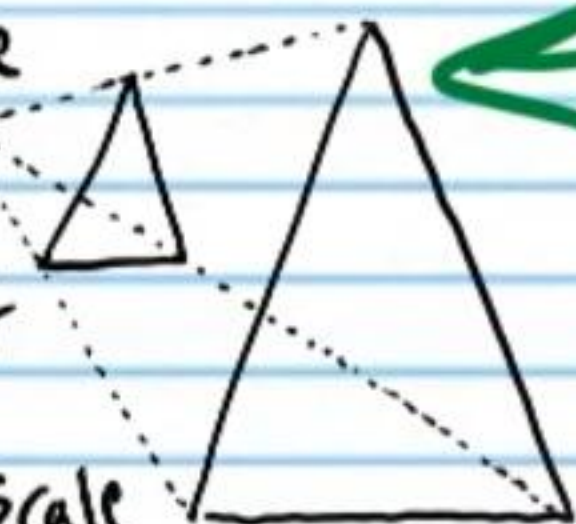
III) Reflection.

* There must be a line of reflection.



IV) Dilation.

* There must be a center of dilation and a scale factor.



No Change
in Size.
Rigid

Size Changes.
Non-Rigid

Relations

Relation: A connection between two values (usually called x and y).

Three Ways to Represent a Relation

Examples

I. Equation

$$y = x^2$$

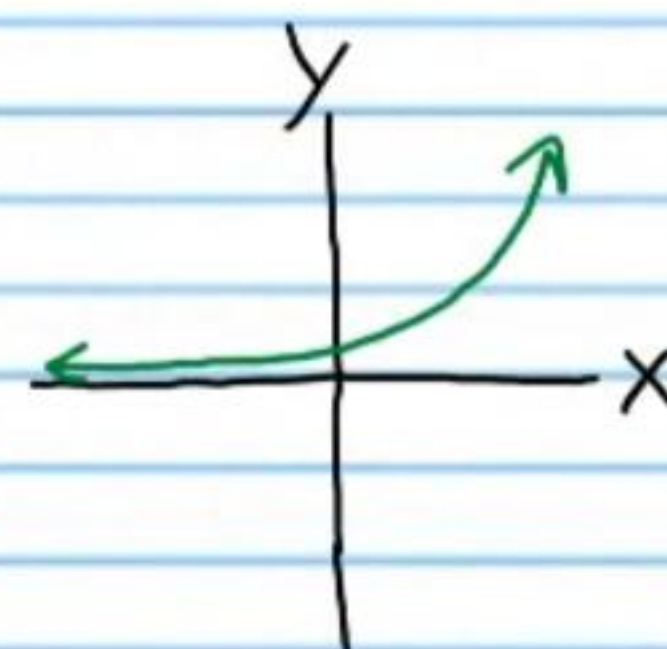
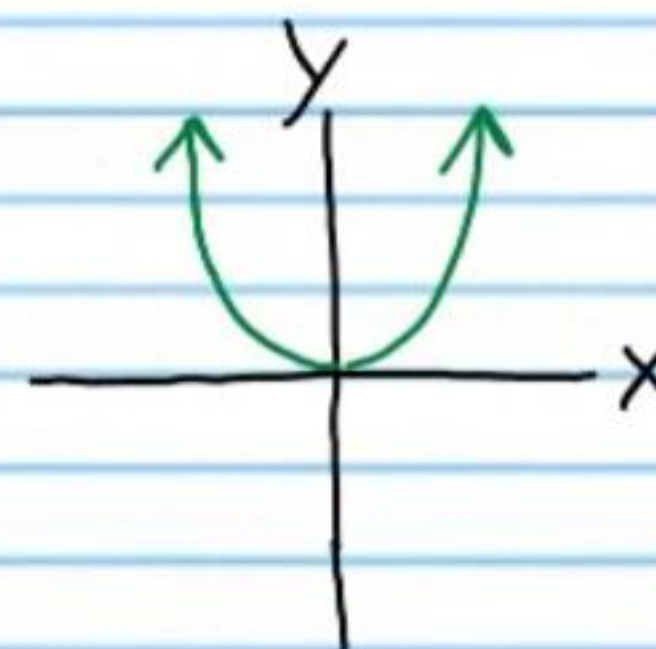
$$y = 3^x$$

II. Table

x	-2	-1	0	1	2
y	4	1	0	1	4

x	-2	-1	0	1	2
y	1/9	1/3	1	3	9

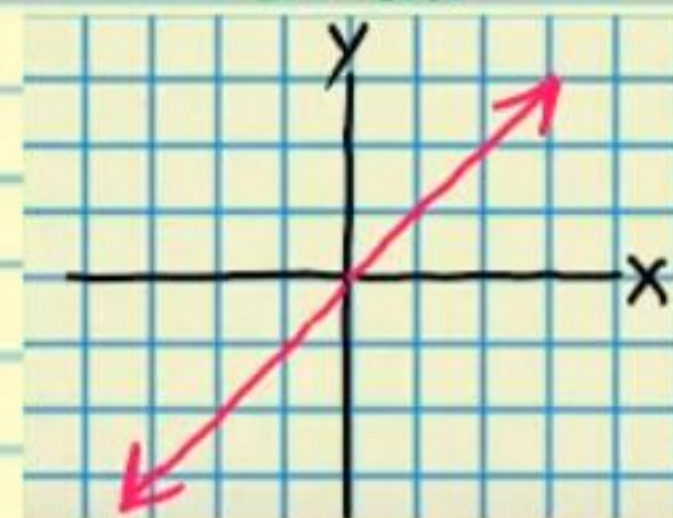
III. Graph



The Library of Relations

* Also called "parent" relations.

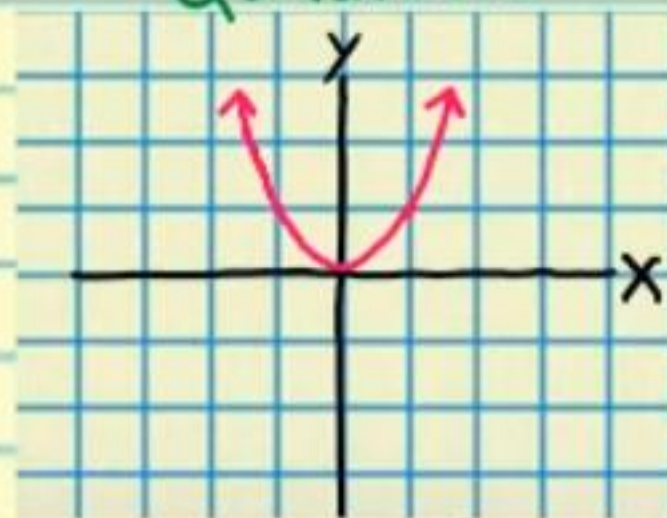
Linear



$$Y = X$$

X	-2	-1	0	1	2
Y	-2	-1	0	1	2

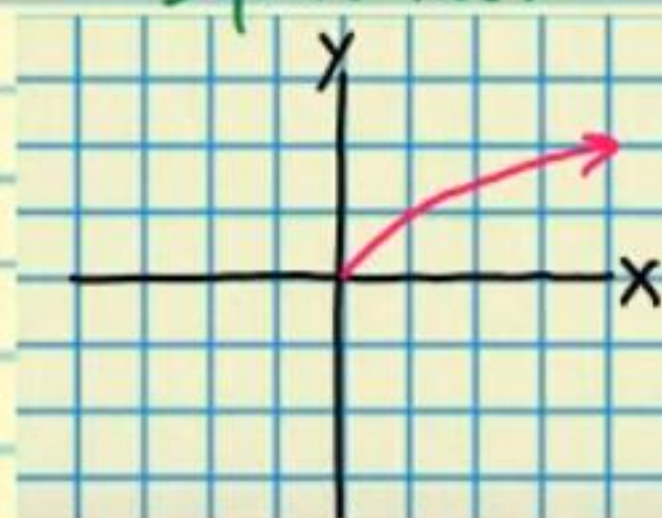
Quadratic



$$Y = x^2$$

X	-2	-1	0	1	2
Y	4	1	0	1	4

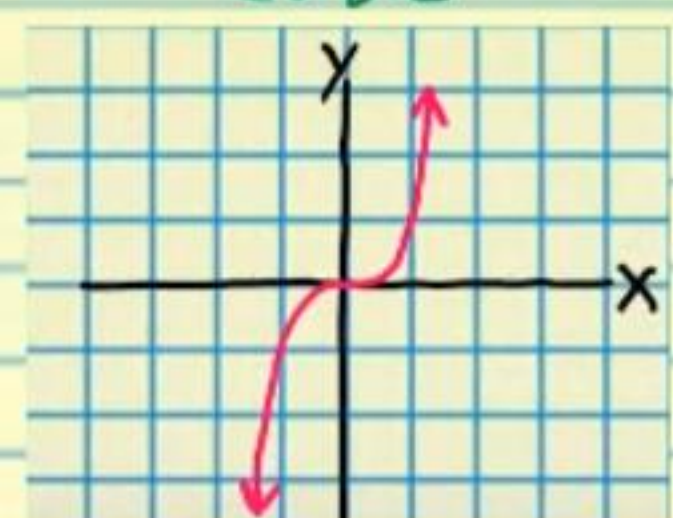
Square Root



$$Y = \sqrt{X}$$

X	0	1	4	9	16
Y	0	1	2	3	4

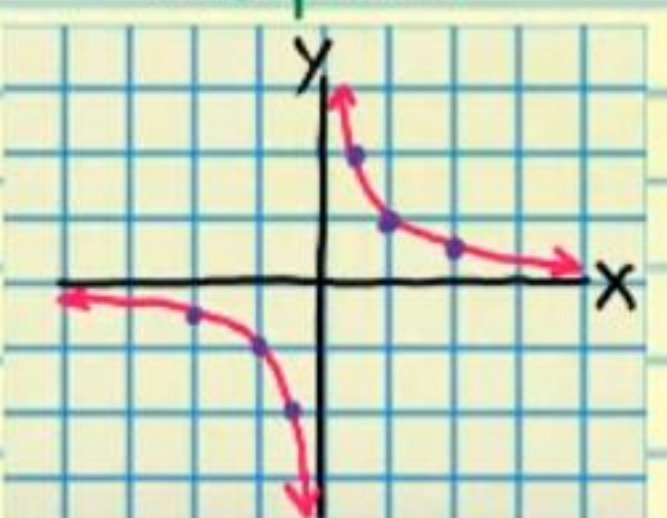
Cubic



$$Y = x^3$$

X	-2	-1	0	1	2
Y	-8	-1	0	1	8

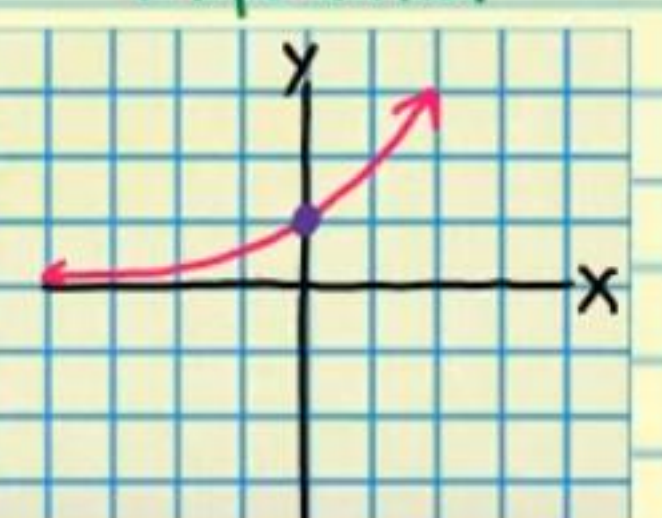
Reciprocal



$$Y = \frac{1}{x}$$

X	-2	-1	-1/2	0	1/2	1	2
Y	-1/2	-1	-2	undefined	2	1	1/2

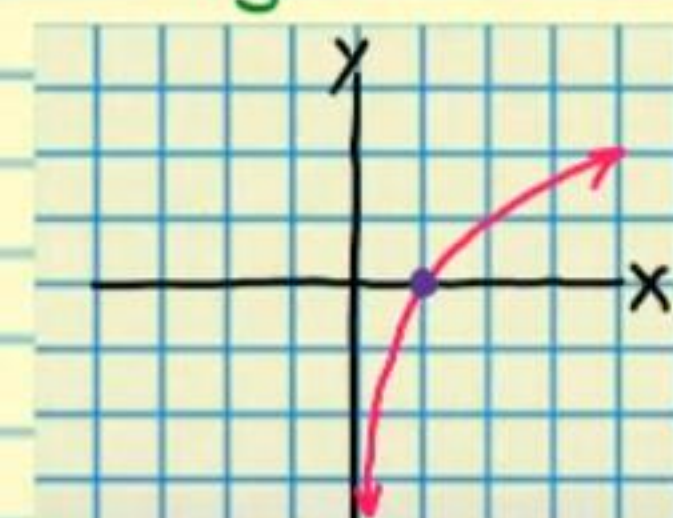
Exponential



$$Y = a^x, a > 1$$

X	-	-	0	-	-
Y	-	-	1	-	-

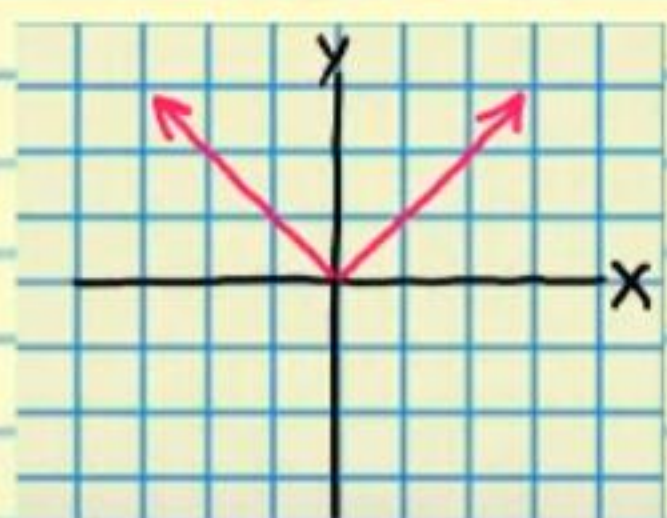
Logarithmic



$$Y = \log_a X$$

X	0	1	-	-	-
Y	undefined	0	-	-	-

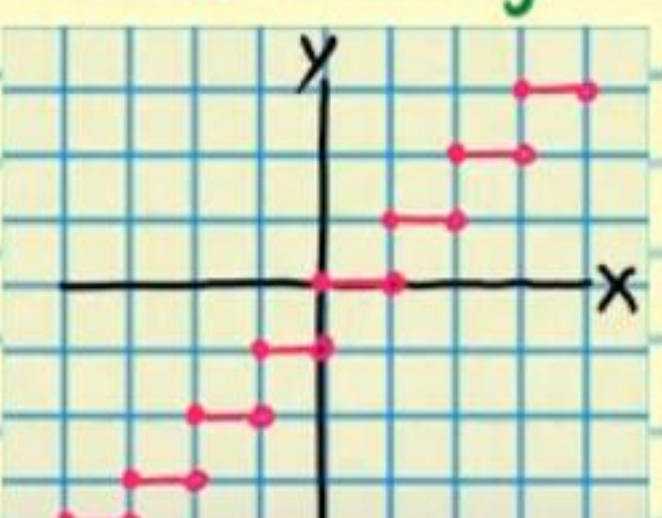
Absolute Value



$$Y = |x|$$

X	-2	-1	0	1	2
Y	2	1	0	1	2

Greatest Integer



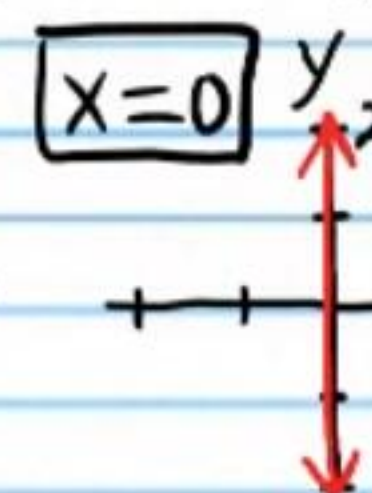
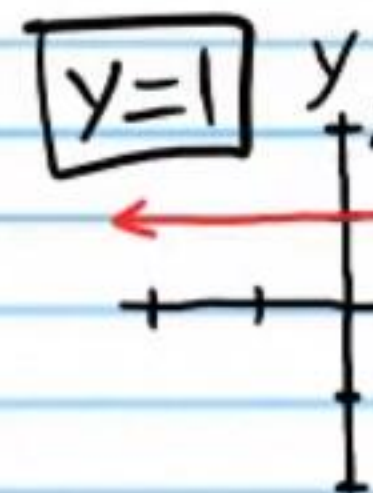
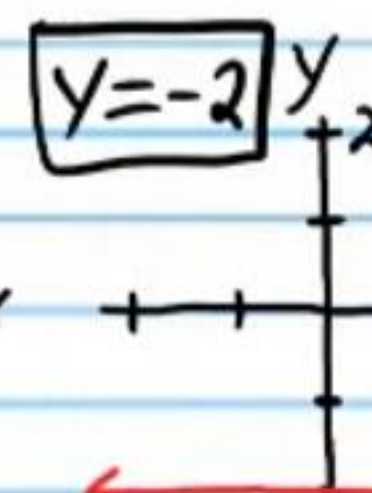
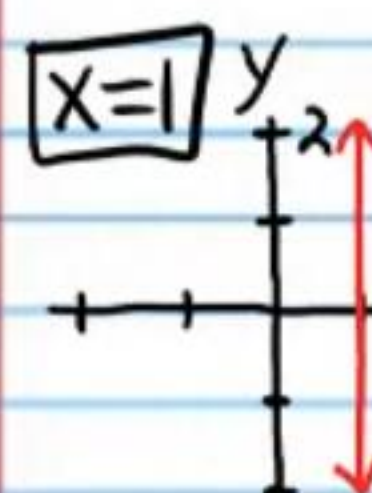
$$Y = [x]$$

X	-2 1/2	-1	0	1/2	1 1/2
Y	-3	-1	0	0	1

Review on Equations of Vertical and Horizontal Lines

Write the equation of each line below.

Example I) Example II) Example III) Example IV)



Translations

Translating a Graph Vertically

If y is changed to $y - k$ in an equation, then its corresponding graph is moved up k units (assuming that k is positive).

If y is changed to $y + k$ in an equation, then its corresponding graph is moved down k units (assuming that k is positive).

Translating a Graph Horizontally

If x is changed to $x - k$ in an equation, then its corresponding graph is moved to the right k units (assuming that k is positive).

If x is changed to $x + k$ in an equation, then its corresponding graph is moved to the left k units (assuming that k is positive).

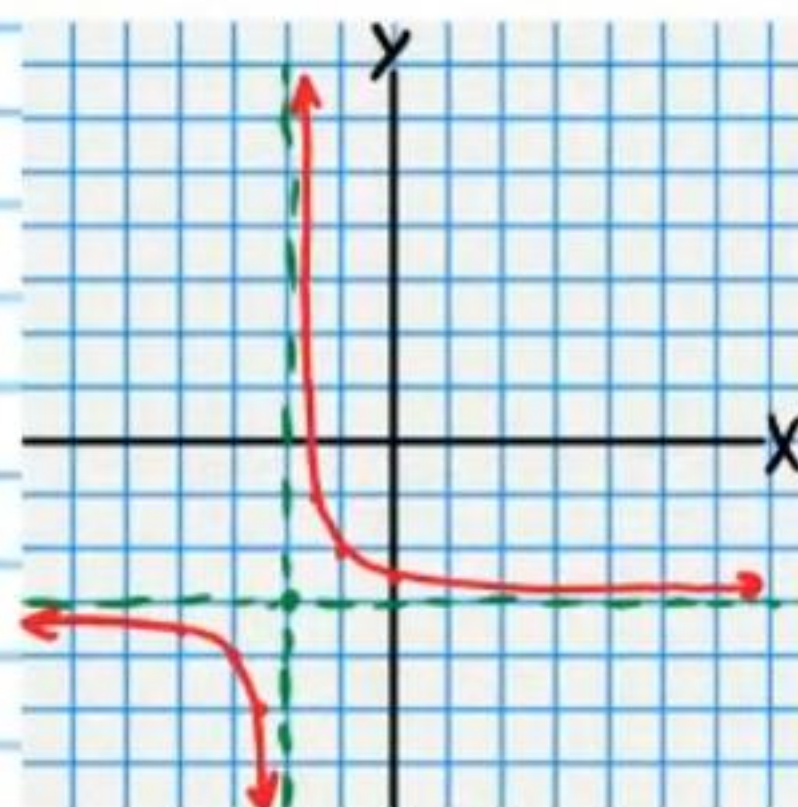
Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted lines and write their equation(s).

① $y+3 = \frac{1}{x+2}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $x = -2, y = -3$

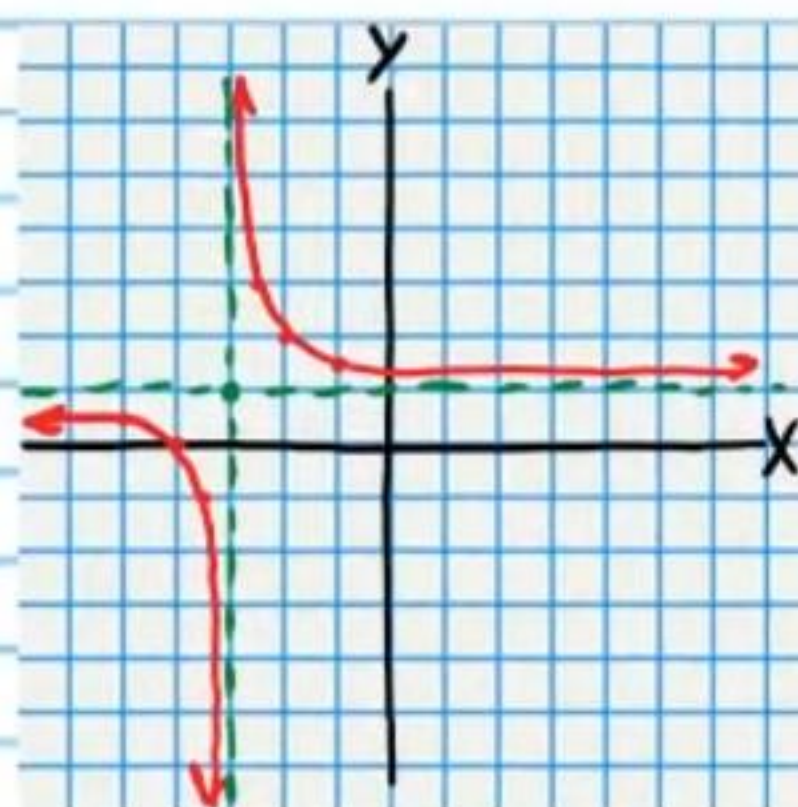


② $\frac{1}{x+3} = y-1$

Parent Relation: Reciprocal

Parent Equation: $\frac{1}{x} = y$

Asymptote Equations: $x = -3, y = 1$

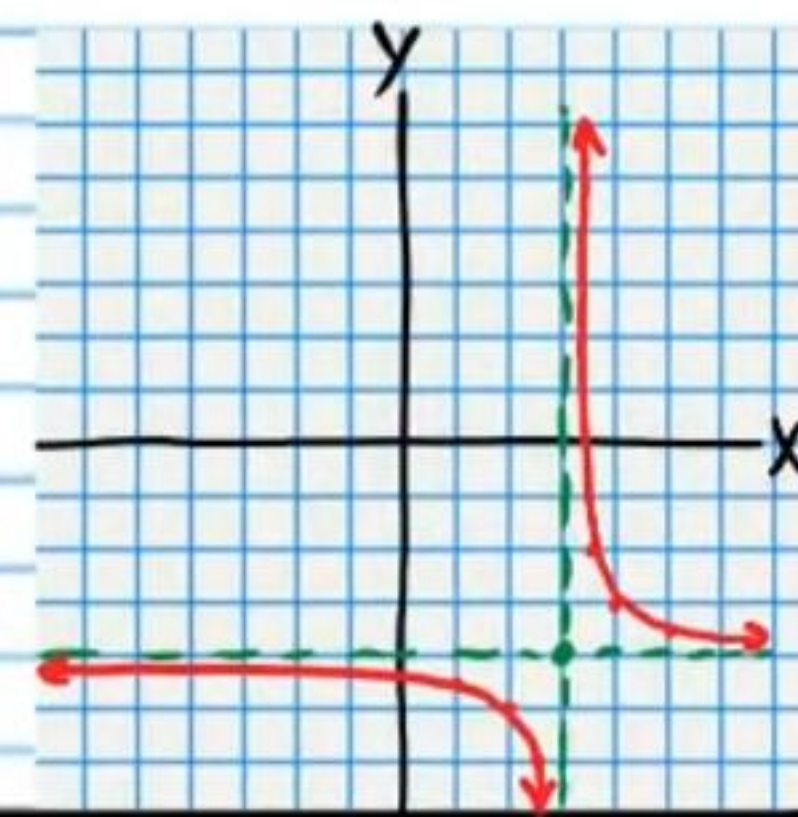


③ $y+4 = \frac{1}{x-3}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $x = 3, y = -4$

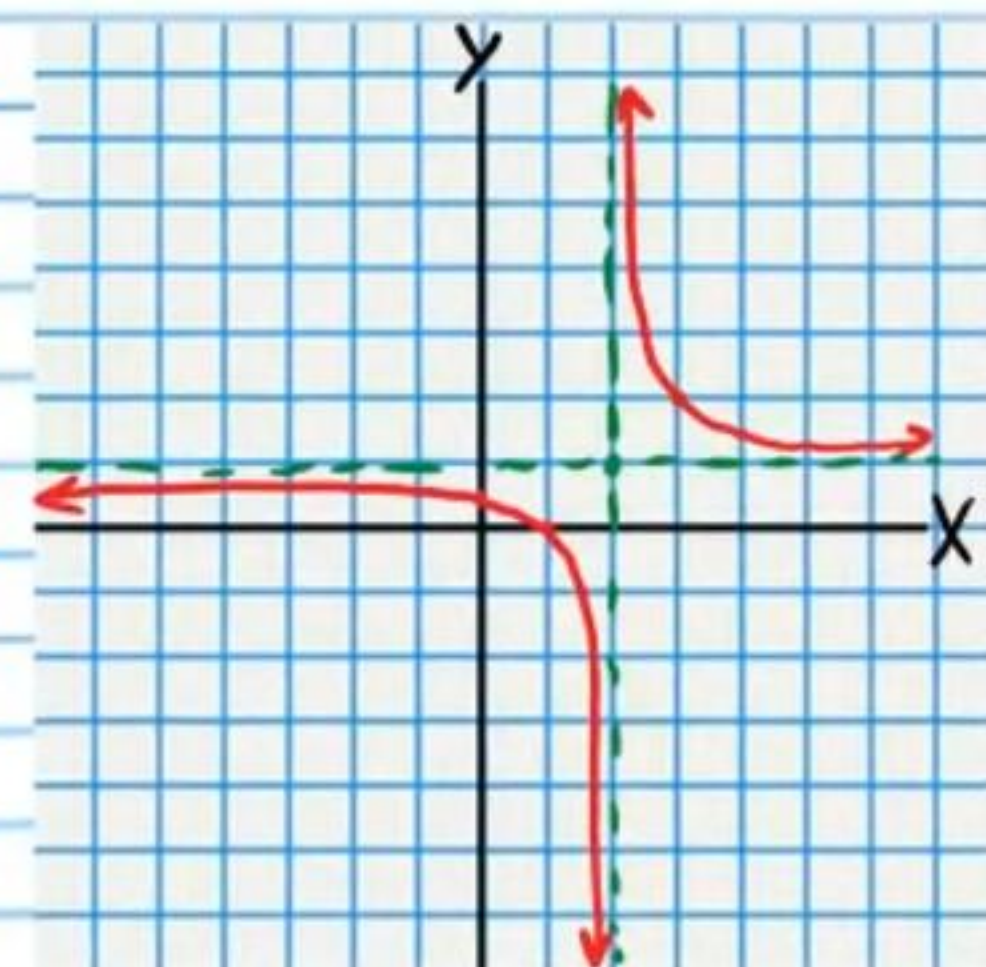


④ $\frac{1}{x-2} = y-1$

Parent Relation: Reciprocal

Parent Equation: $\frac{1}{x} = y$

Asymptote Equations: $x = 2, y = 1$



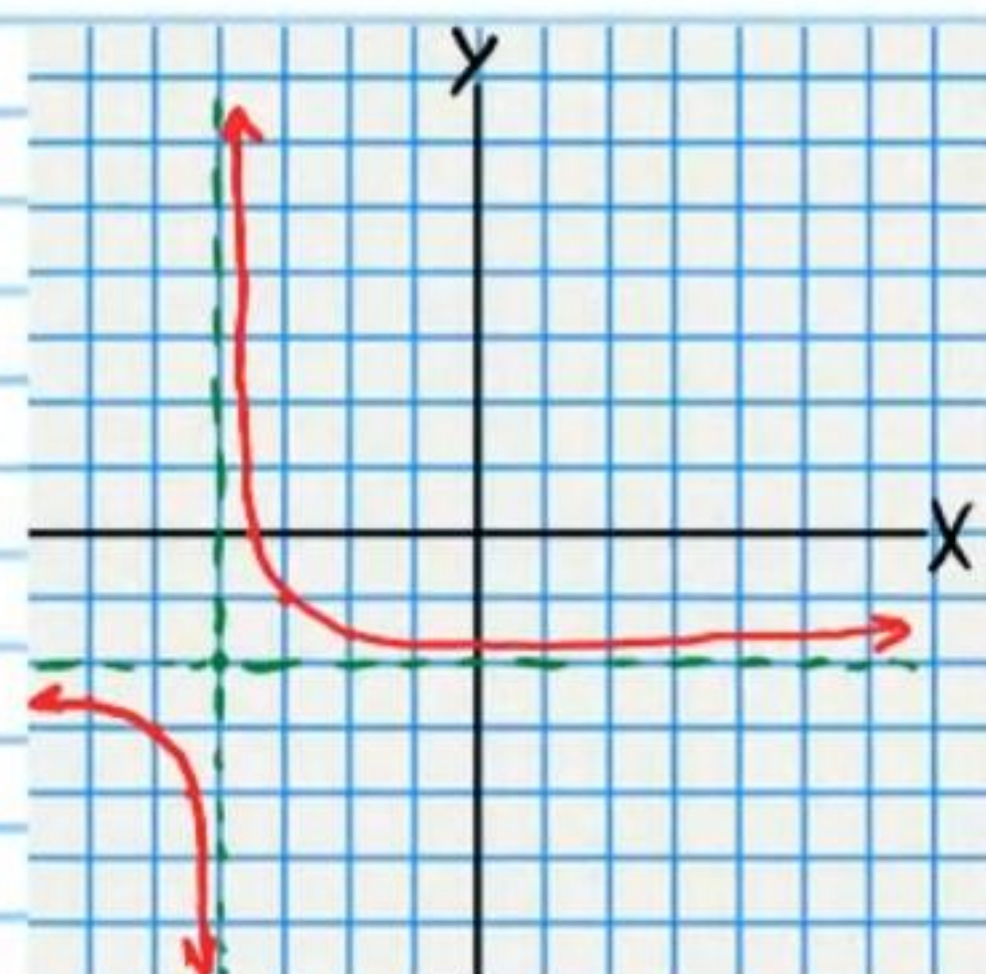
⑤ $y+2 = \frac{1}{x+4}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations:

$x = -4, y = -2$

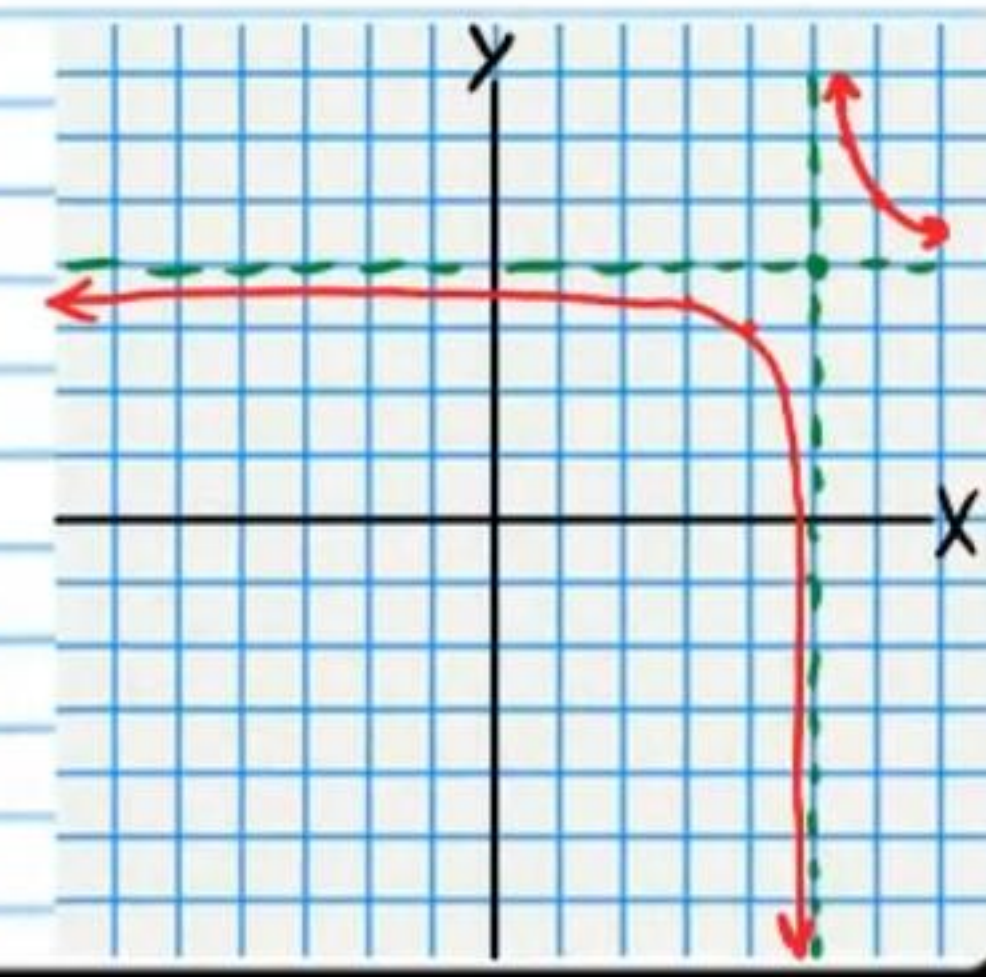


⑥ $y-4 = \frac{1}{x-5}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $x = 5, y = 4$

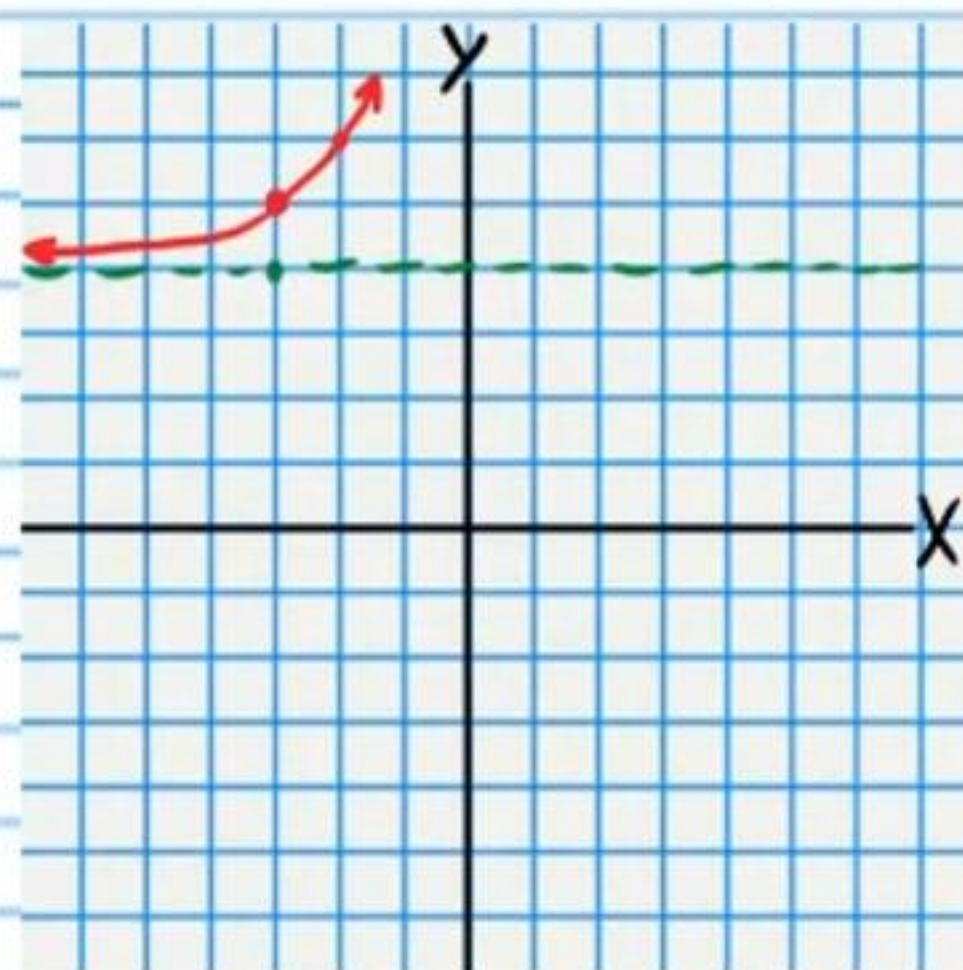


$$\textcircled{7} \quad y-4 = 2^{x+3}$$

Parent Relation: Exponential

Parent Equation: $y = 2^x$

Asymptote Equation: $y = 4$

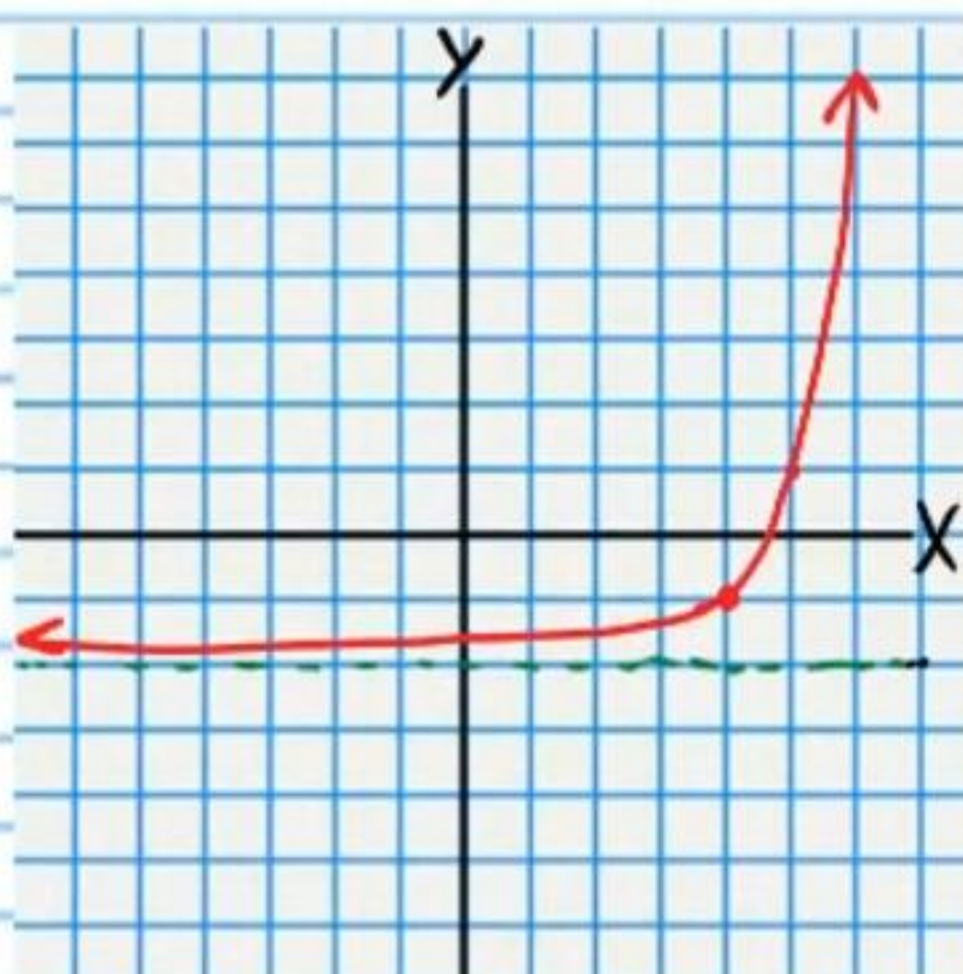


$$\textcircled{8} \quad 3^{x-4} = y+2$$

Parent Relation: Exponential

Parent Equation: $3^x = y$

Asymptote Equation: $y = -2$

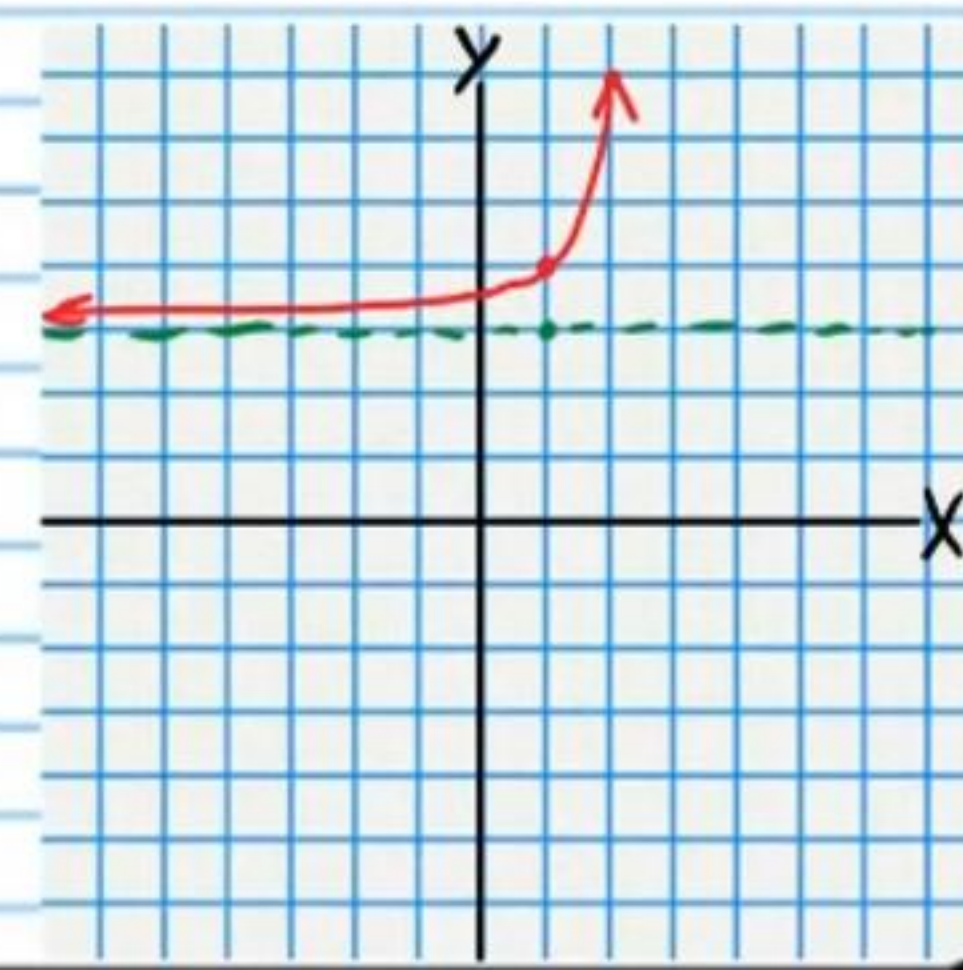


$$\textcircled{9} \quad y-3 = 4^{x-1}$$

Parent Relation: Exponential

Parent Equation: $y = 4^x$

Asymptote Equation: $y = 3$

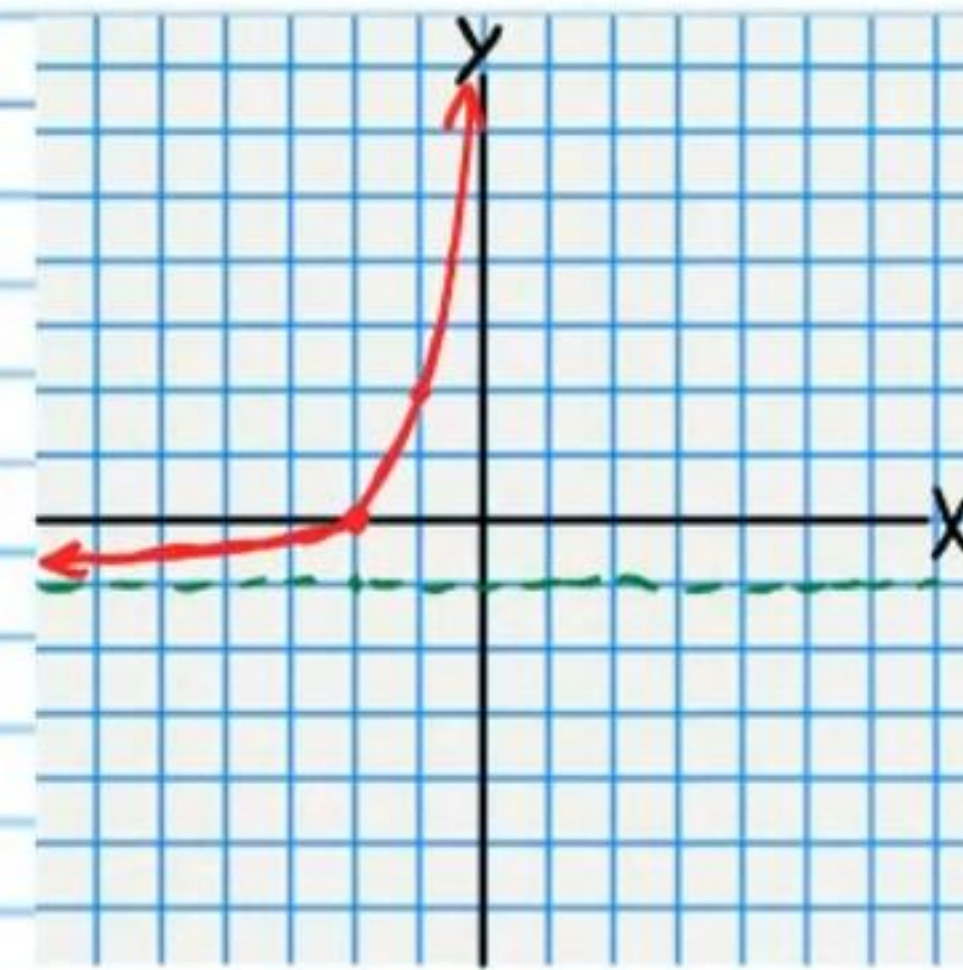


$$\textcircled{10} \quad 3^{x+2} = y+1$$

Parent Relation: Exponential

Parent Equation: $3^x = y$

Asymptote Equation: $y = -1$

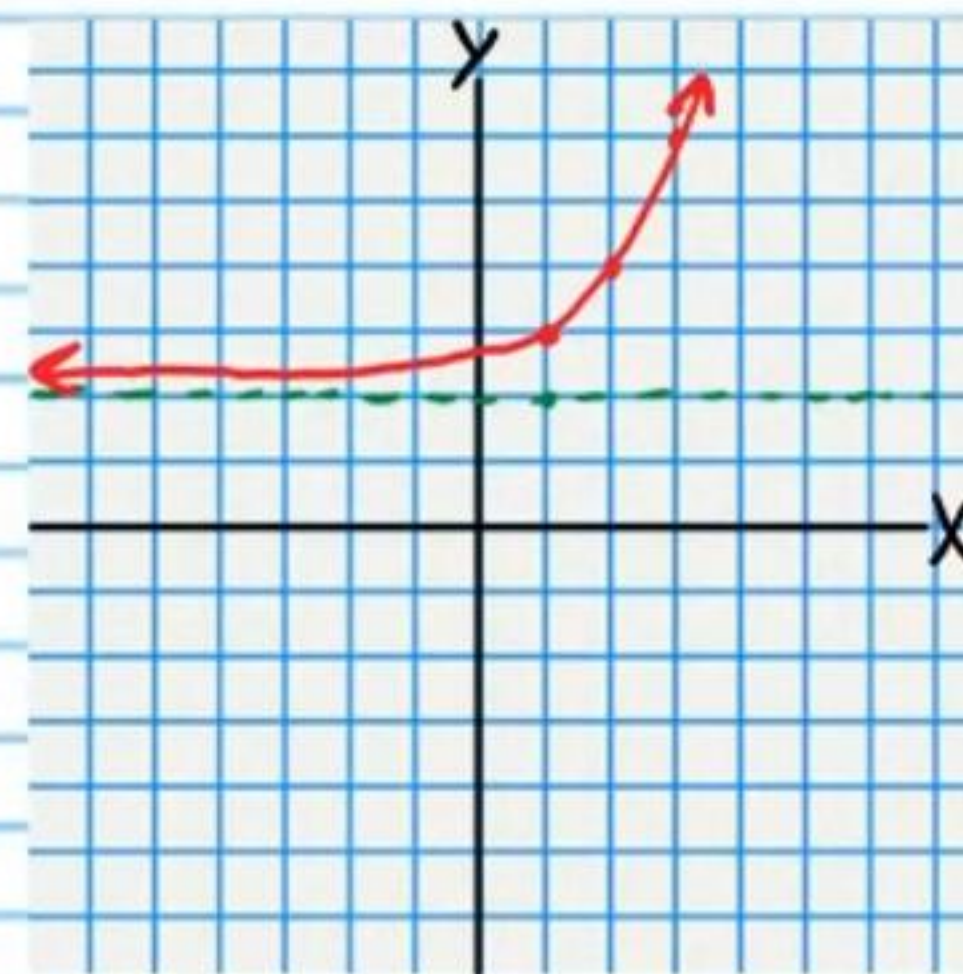


$$\textcircled{11} \quad y-2 = 2^{x-1}$$

Parent Relation: Exponential

Parent Equation: $y = 2^x$

Asymptote Equation: $y = 2$

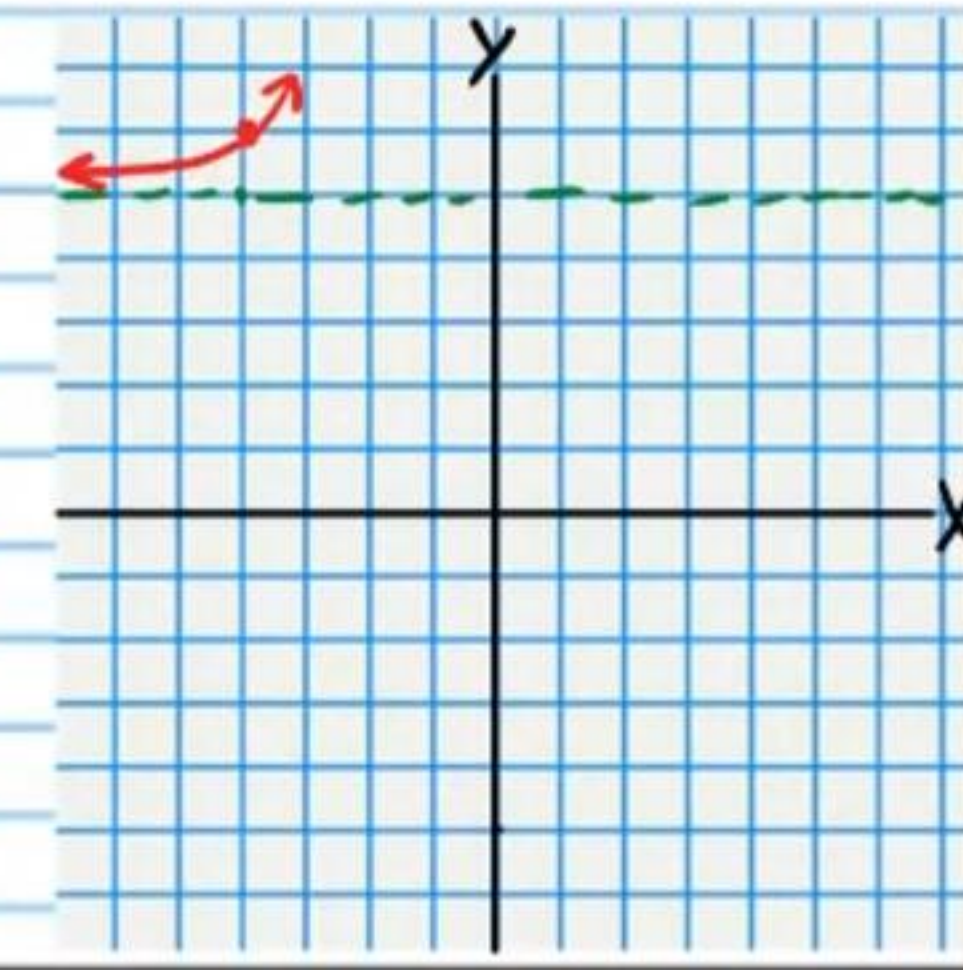


$$\textcircled{12} \quad 3^{x+4} = y-5$$

Parent Relation: Exponential

Parent Equation: $3^x = y$

Asymptote Equation: $y = 5$



Logarithm Review

Find the values of the logarithms below.

Example i) **Logarithm Property III** Example ii) Example iii)

$$\log_2 2$$

$$y = \log_2 2$$

$$2^y = 2$$

$$2^y = 2^1$$

$$y = 1$$

$$\log_2 2 = 1$$

$$\log_b b = 1$$

$$\log_3 3$$

$$\boxed{1}$$

$$\log_6 6$$

$$\boxed{1}$$

Example iv) Example V) Example Vi) **Logarithm Property II**

$$\log_8 8$$

$$\boxed{1}$$

$$\log_{506} 506$$

$$\boxed{1}$$

$$\log_5 25$$

$$\log_5 5^2 = \boxed{2}$$

$$\log_b b^x = x$$

Example vii) Example viii) Example ix) Example x)

$$\log_3 27$$

$$\log_3 3^3 = \boxed{3}$$

$$\log_4 16$$

$$\log_4 4^2 = \boxed{2}$$

$$\log_9 81$$

$$\log_9 9^2 = \boxed{2}$$

$$\log_2 16$$

$$\log_2 2^4 = \boxed{4}$$

Example xi) Example xii) Example xiii) Example xiv)

$$\log_3 1$$

$$\boxed{0}$$

$$\log_7 1$$

$$\boxed{0}$$

$$\log_2 1$$

$$\boxed{0}$$

$$\log_9 1$$

$$\boxed{0}$$

Use These Three Logarithm Properties in the Next Problems

$$\log_b b = 1$$

$$\log_b 1 = 0$$

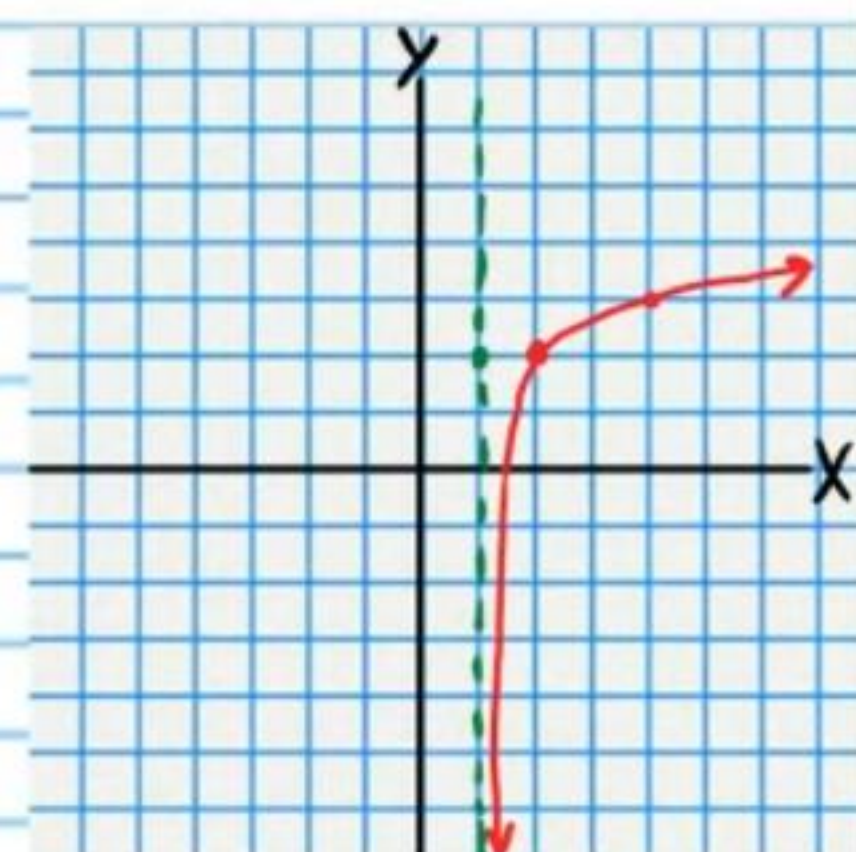
$$\log_b b^x = x$$

$$(13) y - 2 = \log_3 (x - 1)$$

Parent Relation: Logarithmic

Parent Equation: $y = \log_3 x$

Asymptote Equation: $x = 1$

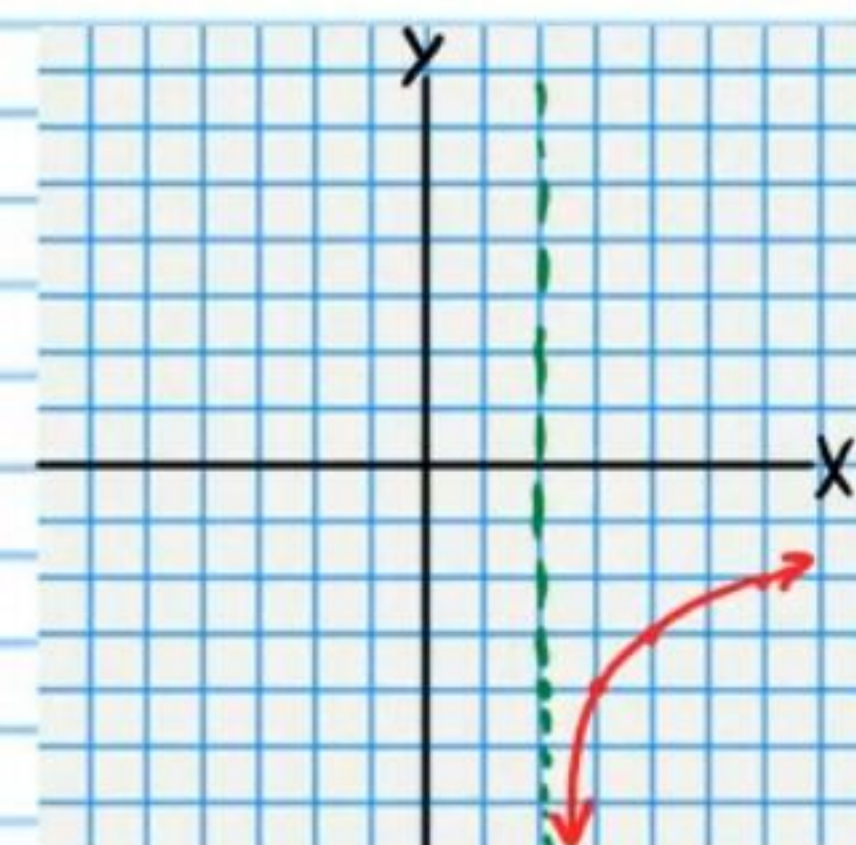


$$(14) \log_2 (x - 2) = y + 4$$

Parent Relation: Logarithmic

Parent Equation: $\log_2 x = y$

Asymptote Equation: $x = 2$

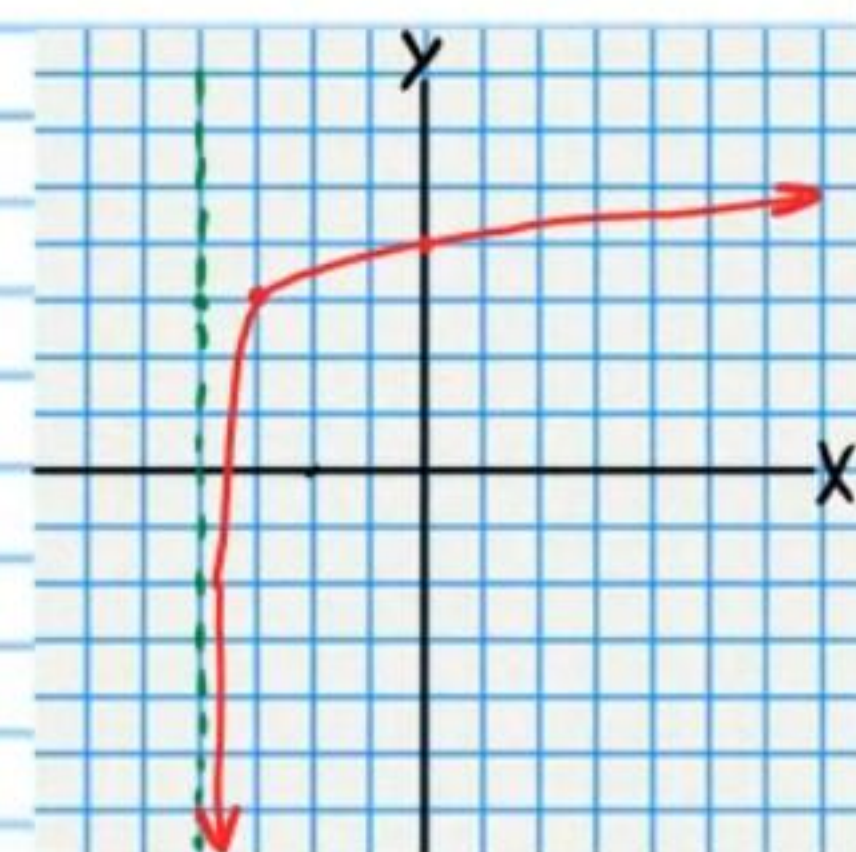


$$(15) y - 3 = \log_4 (x + 4)$$

Parent Relation: Logarithmic

Parent Equation: $y = \log_4 x$

Asymptote Equation: $x = -4$

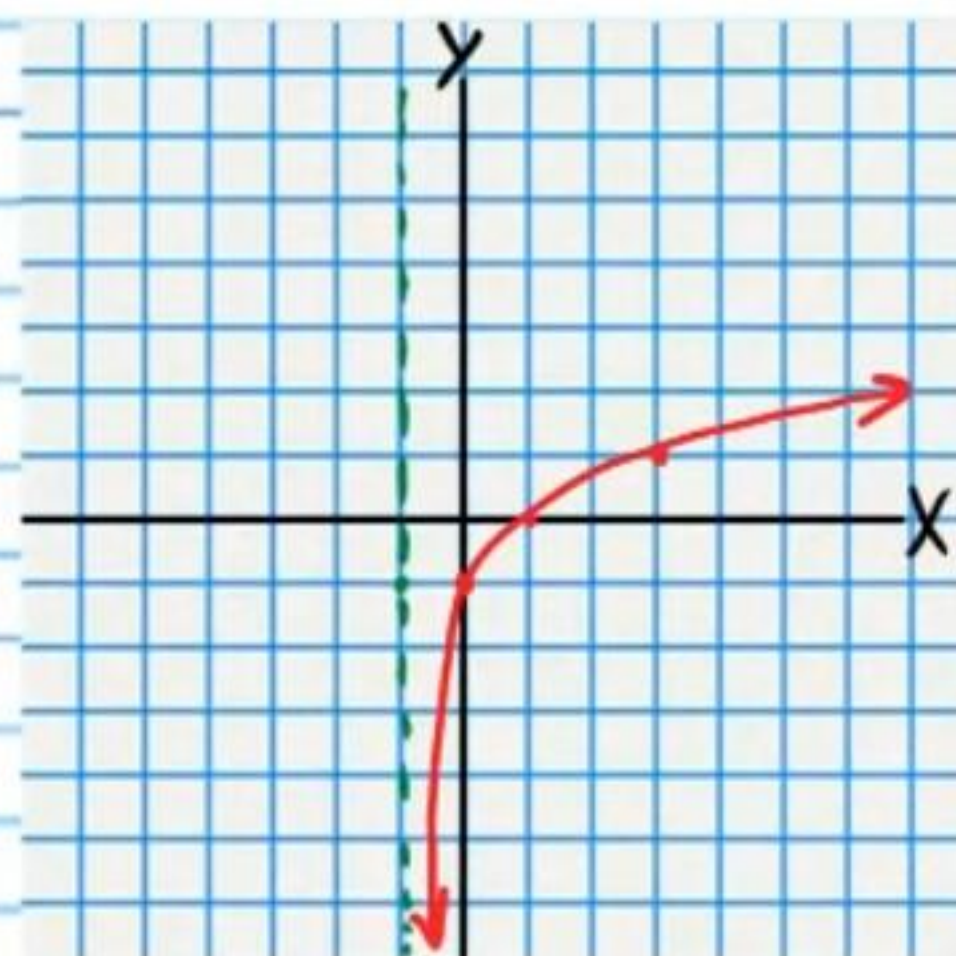


⑩ $\log_2(x+1) = y+1$

Parent Relation: Logarithmic

Parent Equation: $\log_2 x = y$

Asymptote Equation: $x = -1$

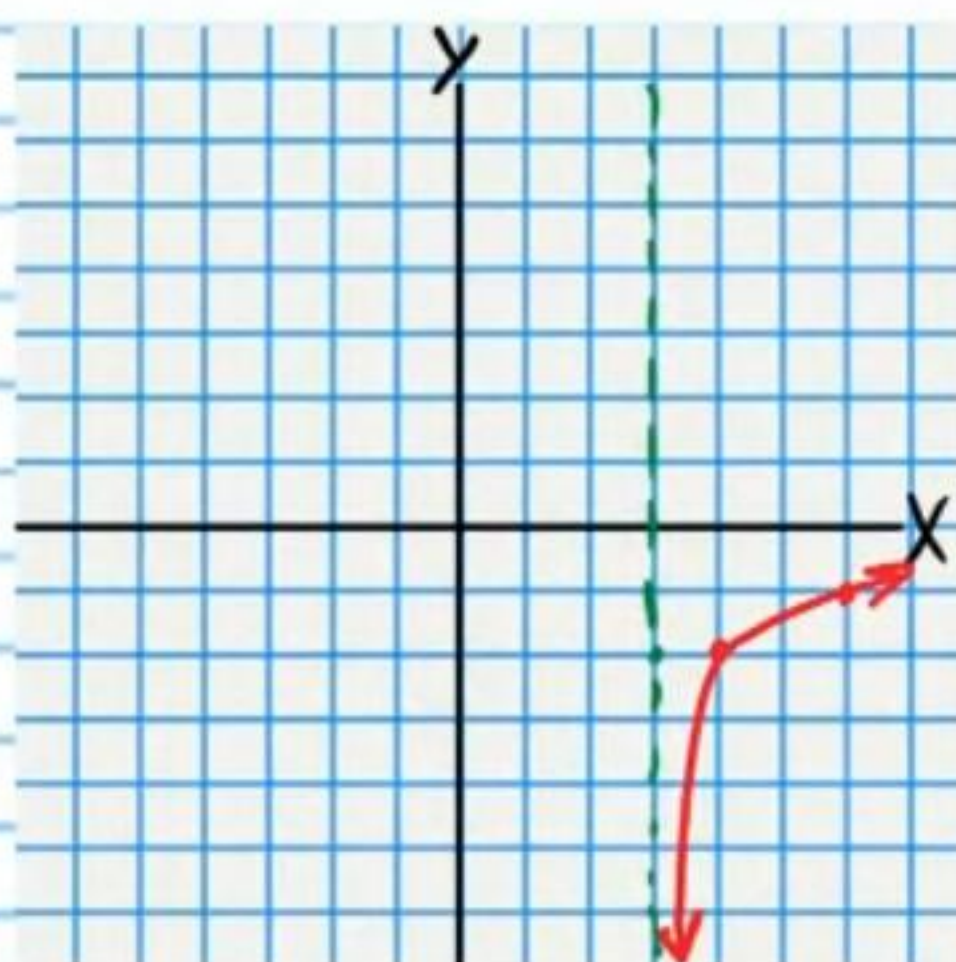


⑪ $\log_3(x-3) = y+2$

Parent Relation: Logarithmic

Parent Equation: $\log_3 x = y$

Asymptote Equation: $x = 3$

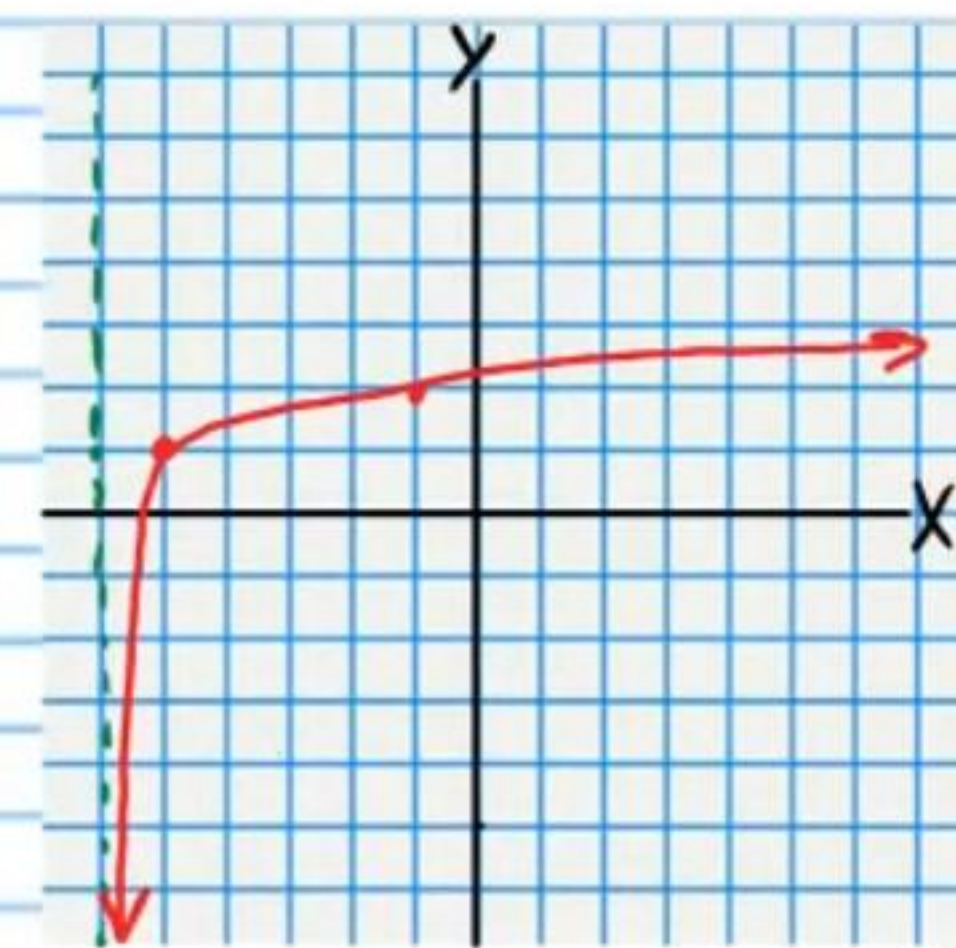


⑫ $y-1 = \log_5(x+6)$

Parent Relation: Logarithmic

Parent Equation: $y = \log_5 x$

Asymptote Equation: $x = -6$



Reflections

Reflecting a Graph Across the X-Axis

If y is changed to $-y$ in an equation, then its corresponding graph is reflected across the x -axis.

Reflecting a Graph Across the Y-Axis

If x is changed to $-x$ in an equation, then its corresponding graph is reflected across the y -axis.

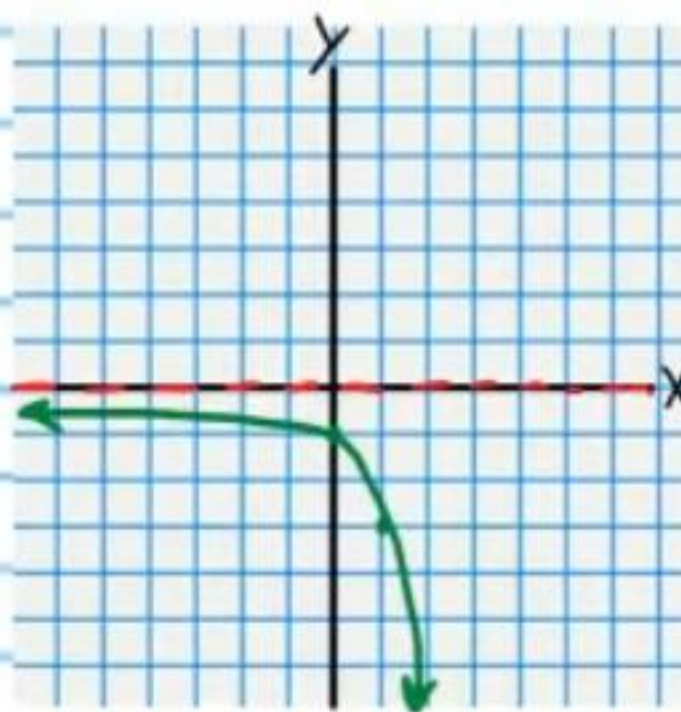
Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted lines and write their equation(s).

⑬ $-y = 3^x$

Parent Relation: Exponential

Parent Equation: $y = 3^x$

Asymptote Equation: $y = 0$

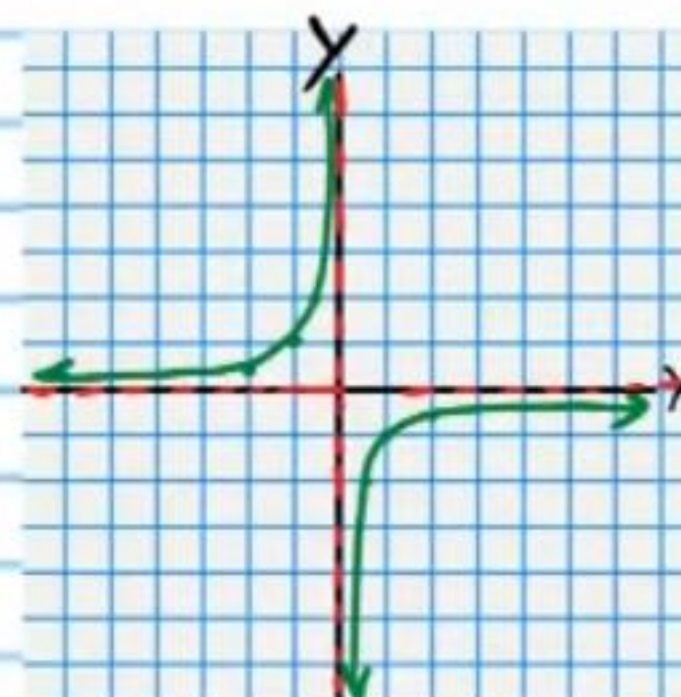


⑭ $y = \frac{1}{-x}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $x = 0, y = 0$

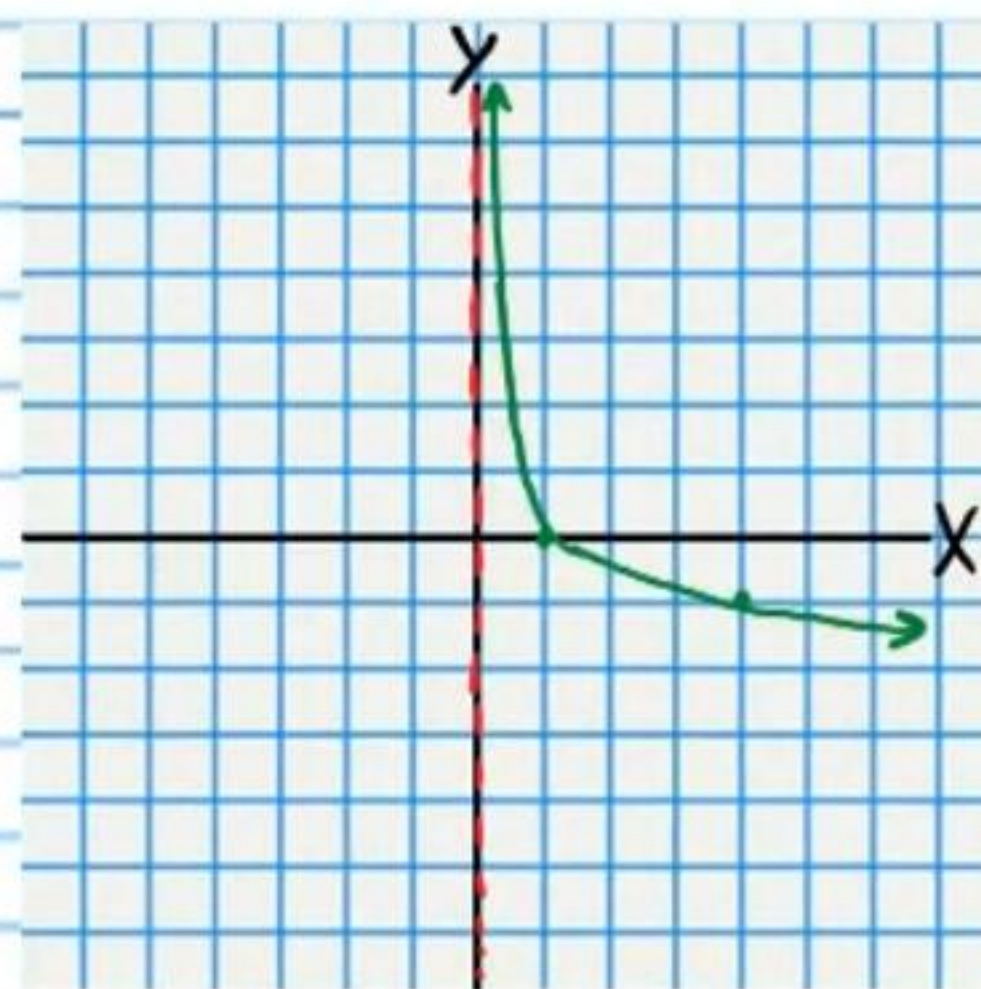


②① $-y = \log_4 x$

Parent Relation: Logarithmic

Parent Equation: $y = \log_4 x$

Asymptote Equation: $x = 0$

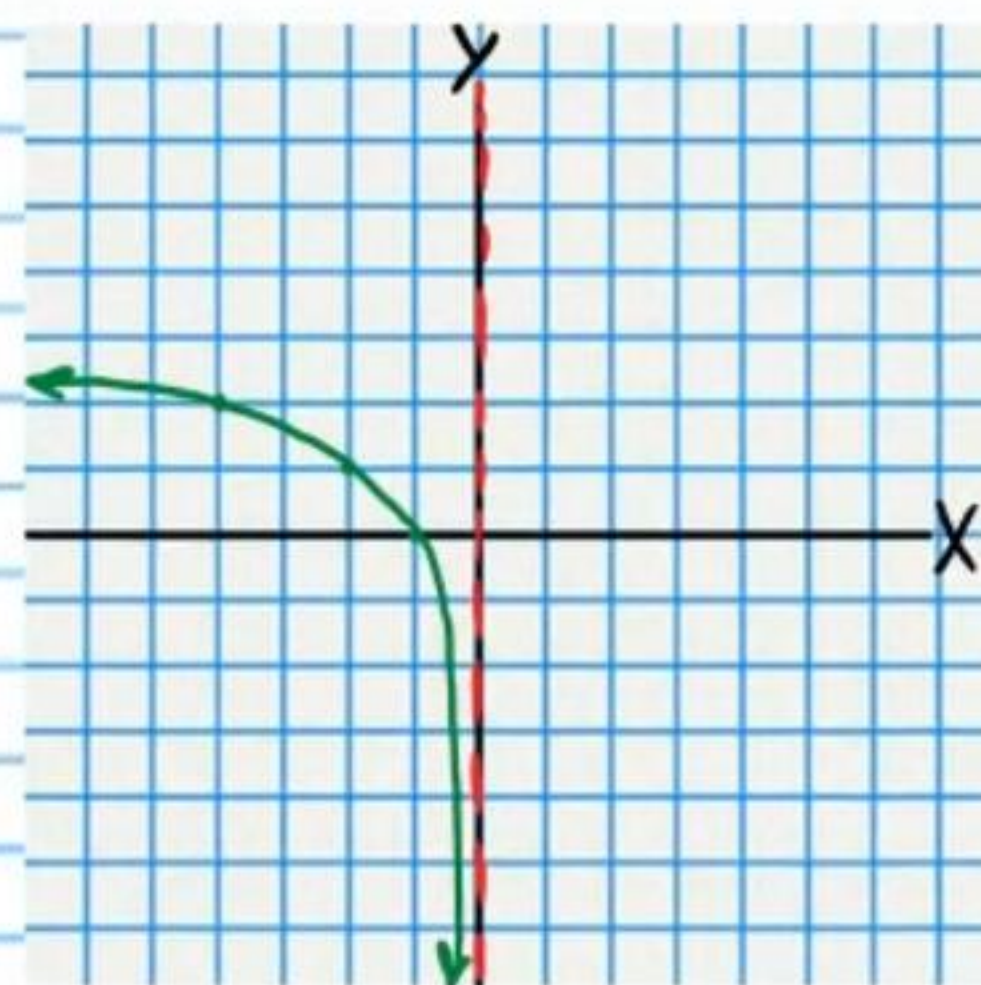


②② $y = \log_2(-x)$

Parent Relation: Logarithmic

Parent Equation: $y = \log_2 x$

Asymptote Equation: $x = 0$

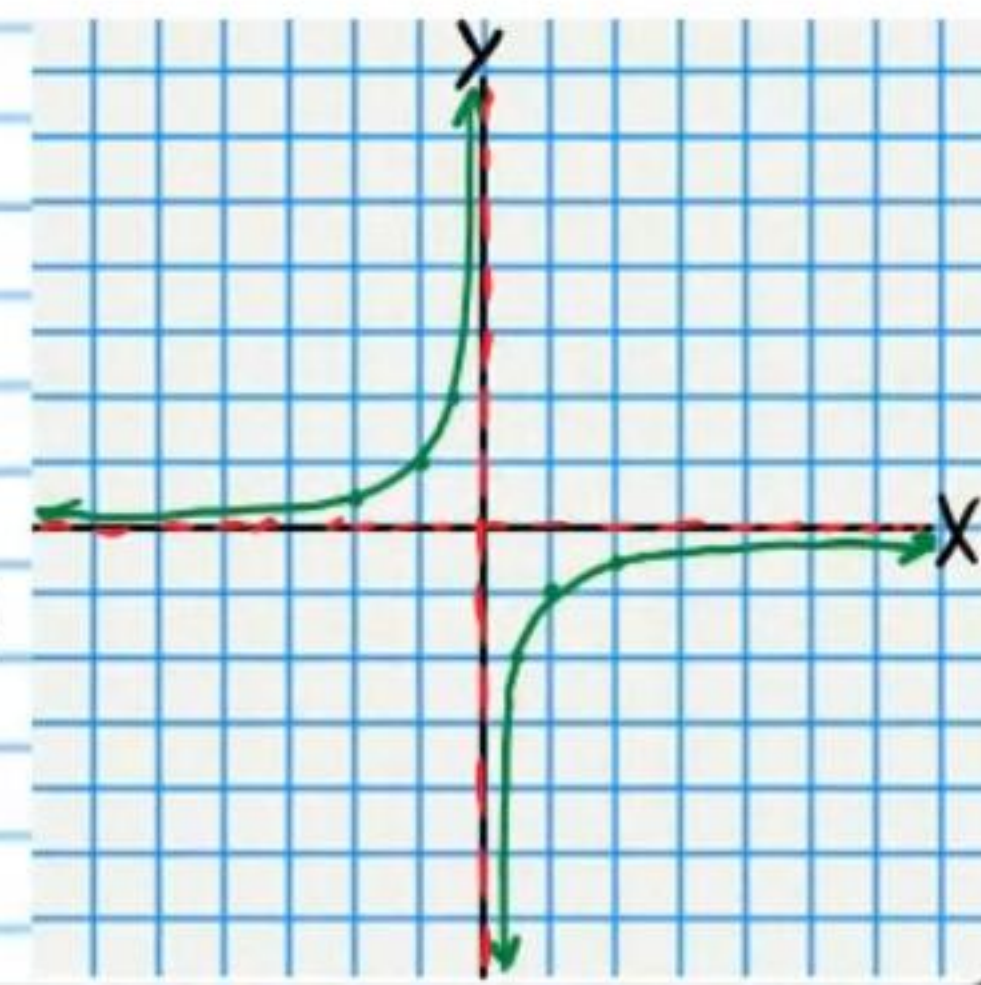


②③ $-y = \frac{1}{x}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equation: $x = 0, y = 0$

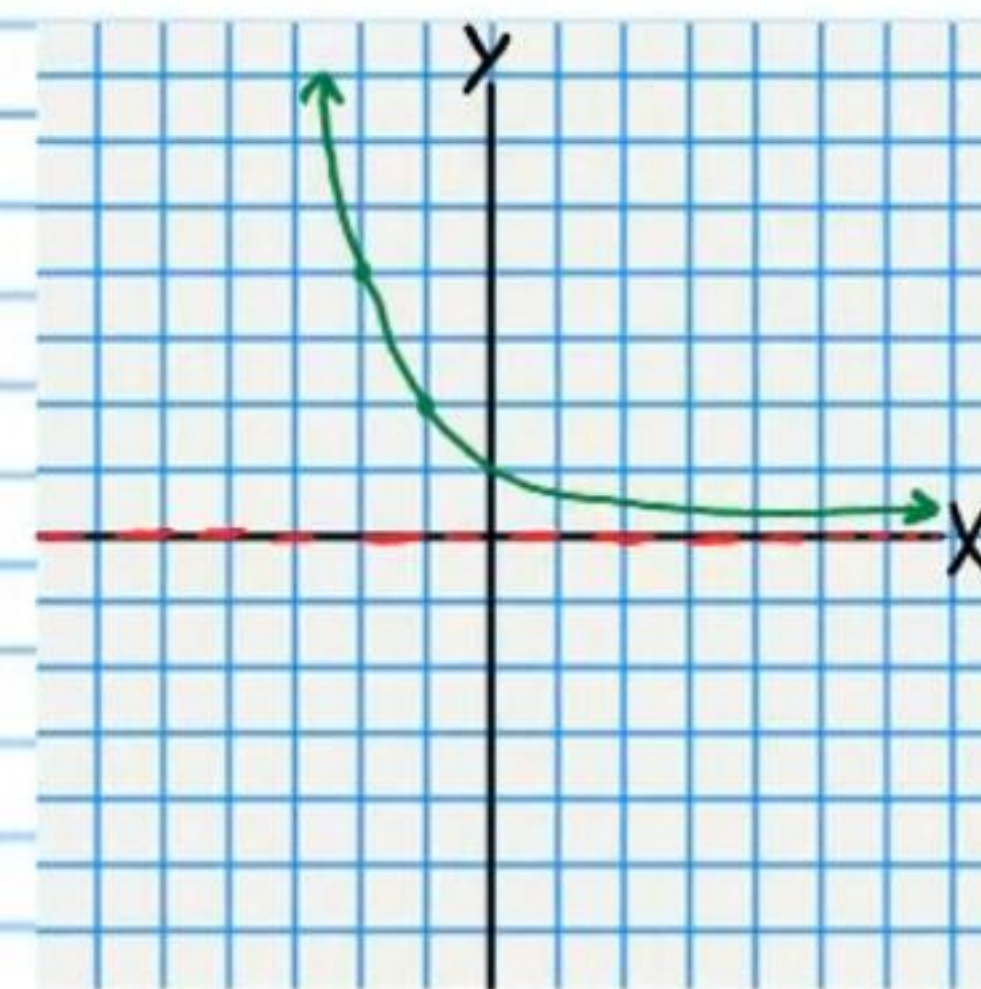


②④ $y = 2^{-x}$

Parent Relation: Exponential

Parent Equation: $y = 2^x$

Asymptote Equations: $y = 0$



Dilations (Compressing and Stretching)

Dilating a Graph Vertically

If y is multiplied by a number in an equation, then its corresponding graph is compressed vertically.

If y is divided by a number in an equation, then its corresponding graph is stretched vertically.

Dilating a Graph Horizontally

If x is multiplied by a number in an equation, then its corresponding graph is compressed horizontally.

If x is divided by a number in an equation, then its corresponding graph is stretched horizontally.

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted line and write their equation(s). Identify the type of dilation involved.

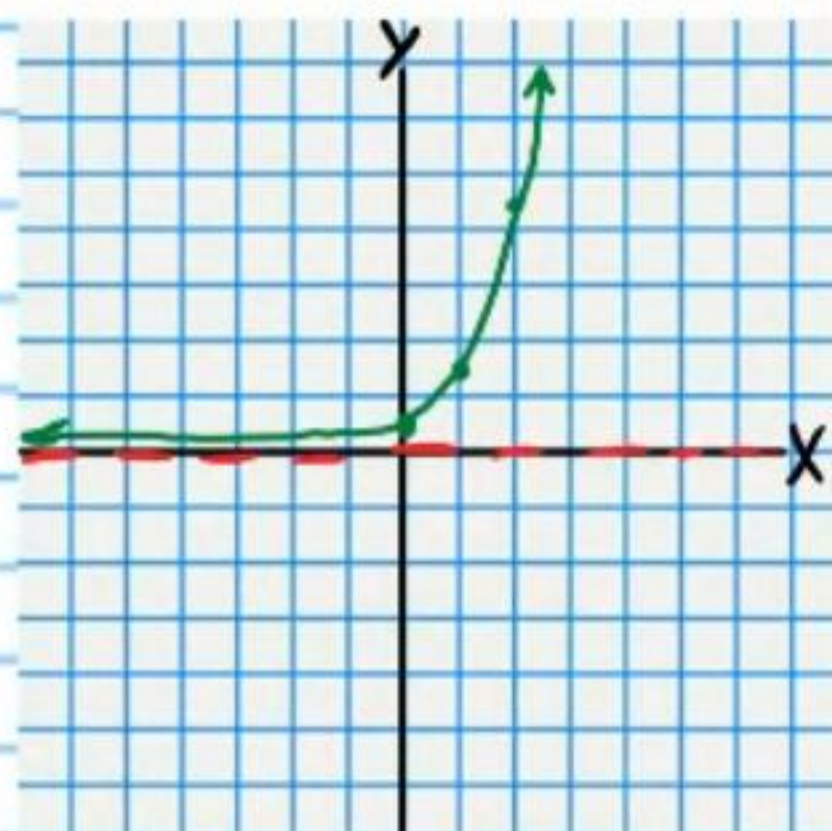
②⑤ $2y = 3^x$

Parent Relation: Exponential

Parent Equation: $y = 3^x$

Asymptote Equation: $y = 0$

Dilation: Vertical Compression



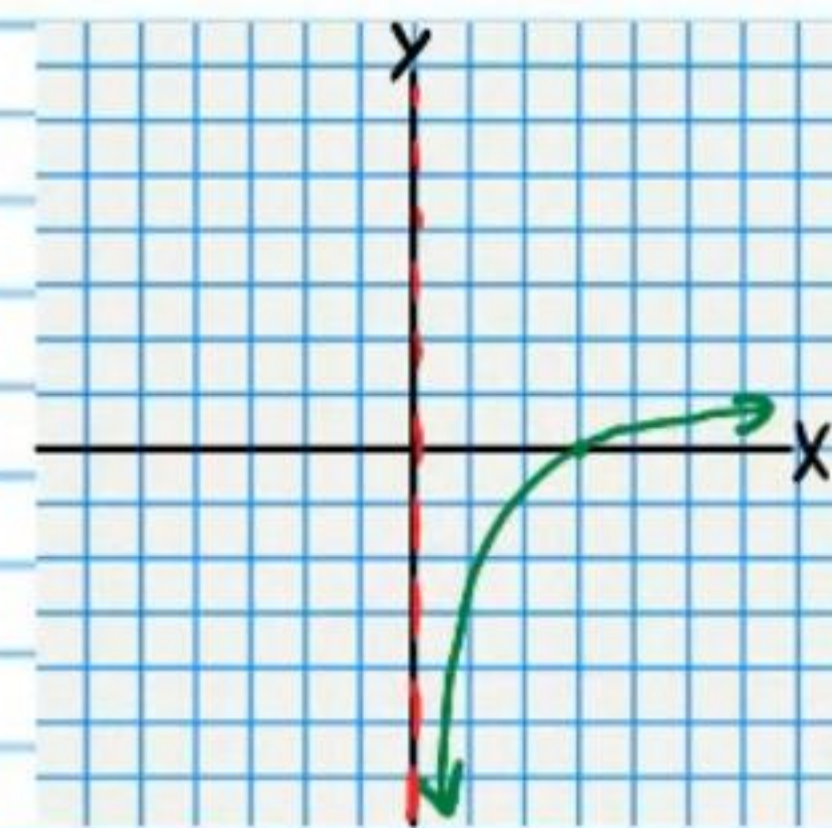
②⑥ $y = \log_4\left(\frac{x}{3}\right)$

Parent Relation: Logarithmic

Parent Equation: $y = \log_4 x$

Asymptote Equation: $x = 0$

Dilation: Horizontal Stretch



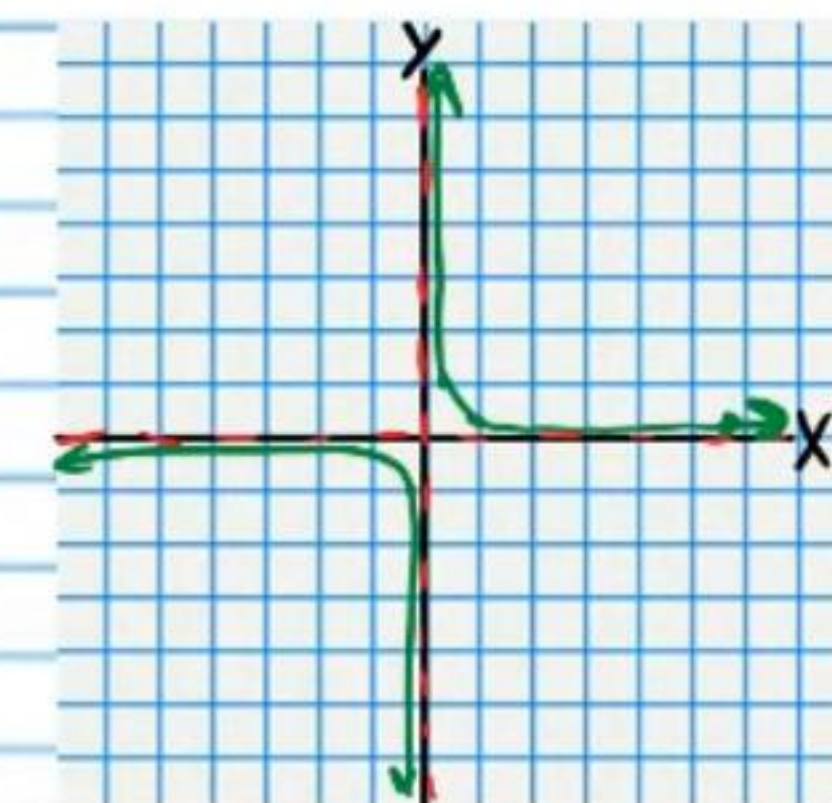
②⑦ $5y = \frac{1}{x}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $x = 0, y = 0$

Dilation: Vertical Compression



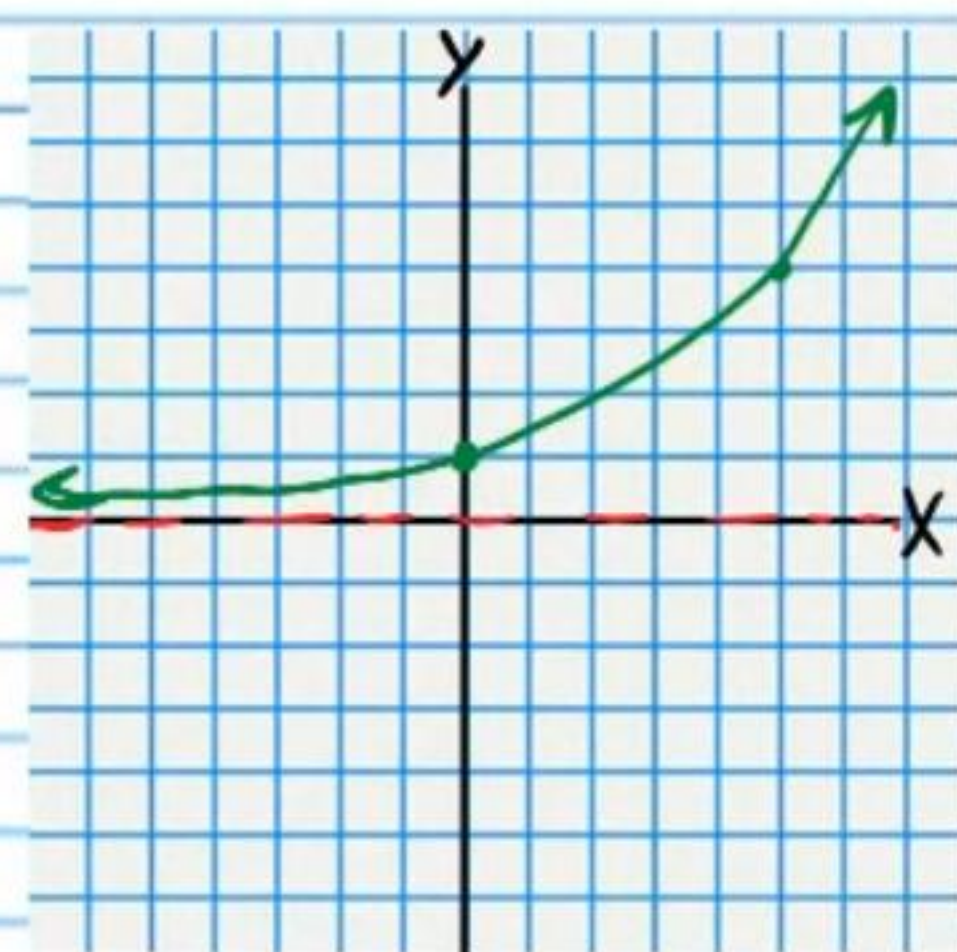
②⑧ $y = 4^{\frac{x}{5}}$

Parent Relation: Exponential

Parent Equation: $y = 4^x$

Asymptote Equation: $y = 0$

Dilation: Horizontal Stretch



②⑨ $\frac{y}{4} = \log_3 x$

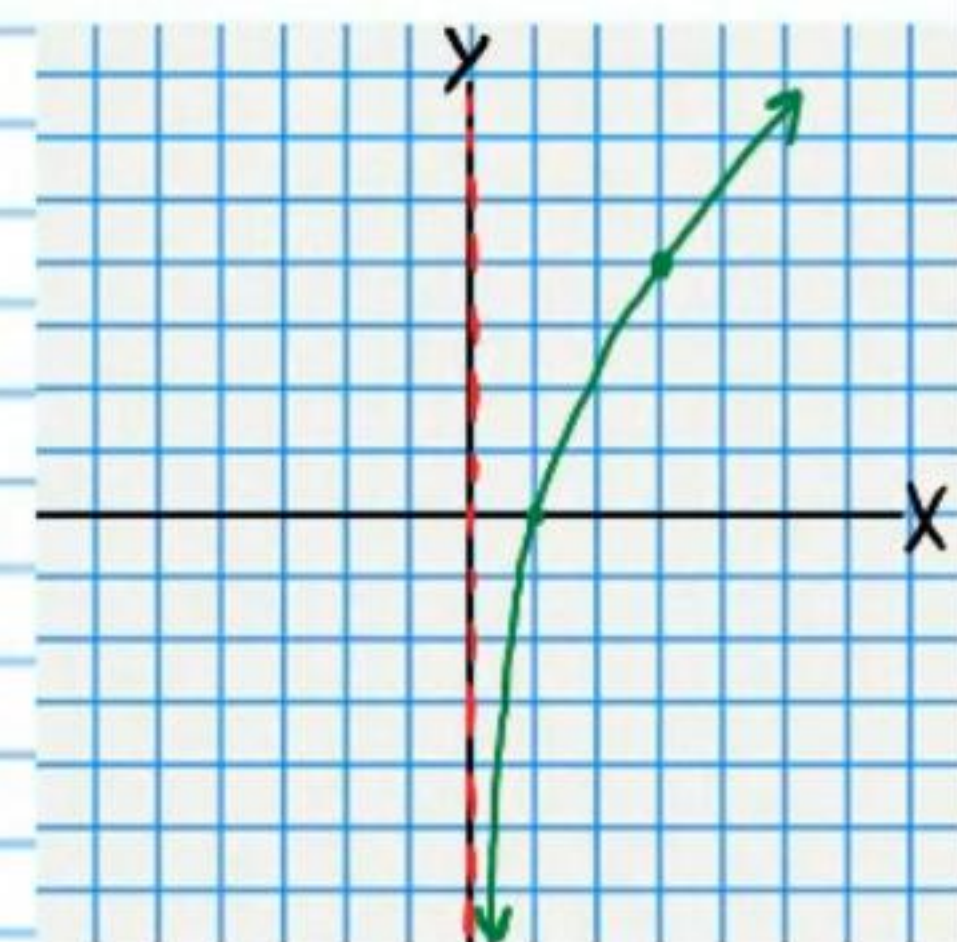
Choose values wisely.

Parent Relation: Logarithmic

Parent Equation: $y = \log_3 x$

Asymptote Equation: $x = 0$

Dilation: Vertical Stretch



③⑩ $y = \frac{1}{2x}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $y = 0, x = 0$

Dilation: Horizontal Compression

